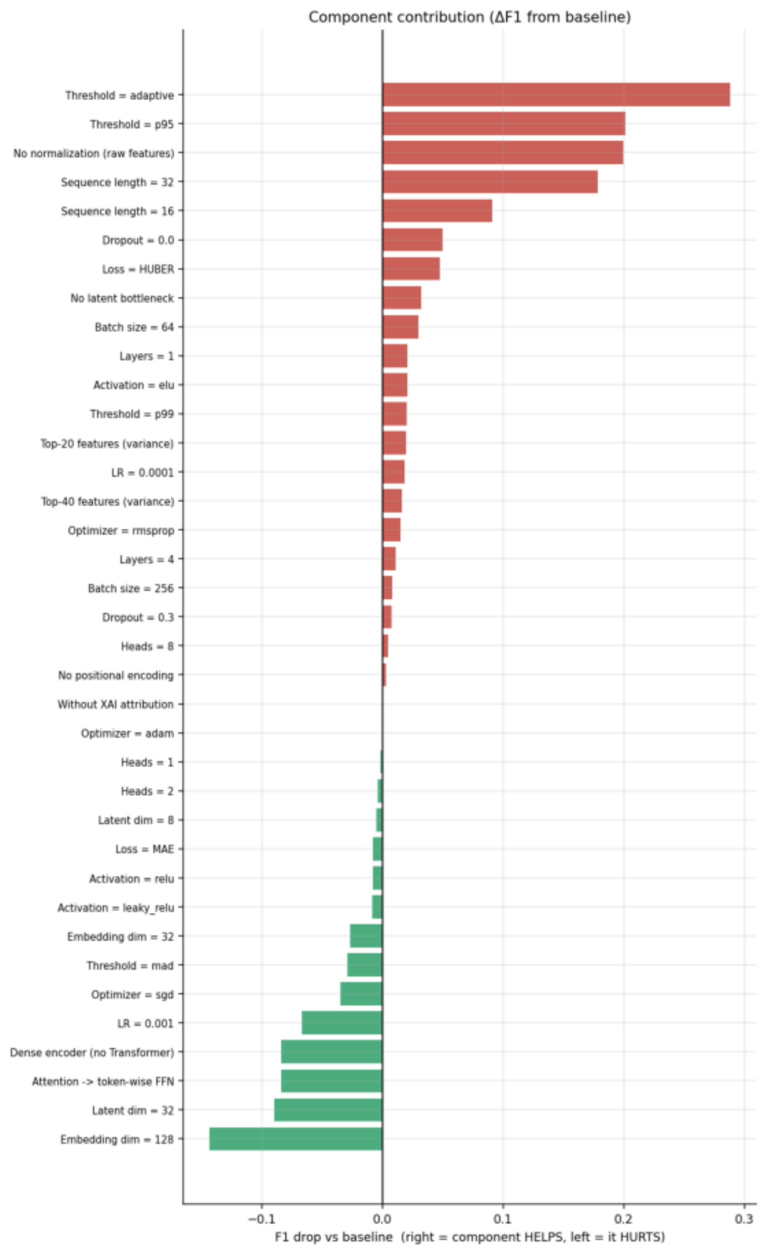
Ablation & Contribution Study

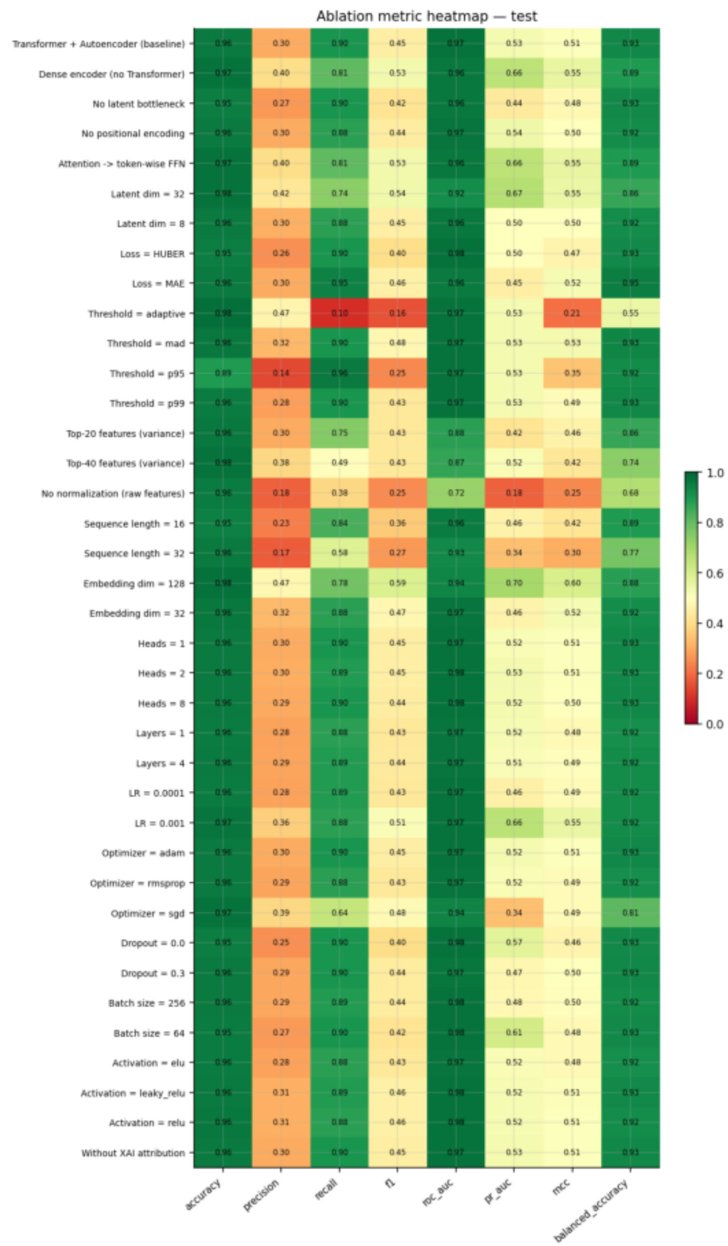
Transformer Autoencoder · leak-free · held-out test

Component	MaxΔF1	Worst variant
Threshold strategy	0.2884	Threshold = adaptive
Normalization	0.1997	No normalization (raw features)
Sequence length	0.1782	Sequence length = 32
Dropout	0.0499	Dropout = 0.0
Reconstruction loss	0.0475	Loss = HUBER
Autoencoder bottleneck	0.0324	No latent bottleneck
Batch size	0.0297	Batch size = 64
Transformer layers	0.0208	Layers = 1
Activation	0.0208	Activation = elu
Feature selection	0.0194	Top-20 features (variance)
Learning rate	0.0184	LR = 0.0001
Optimizer	0.015	Optimizer = rmsprop
Attention heads	0.0045	Heads = 8
Positional encoding	0.003	No positional encoding
XAI (Integrated Gradients)	0.0	Without XAI attribution
Latent dim	-0.0048	Latent dim = 8
Embedding size	-0.0266	Embedding dim = 32
Transformer encoder	-0.0841	Dense encoder (no Transformer)
Self-attention	-0.0841	Attention -> token-wise FFN

contribution



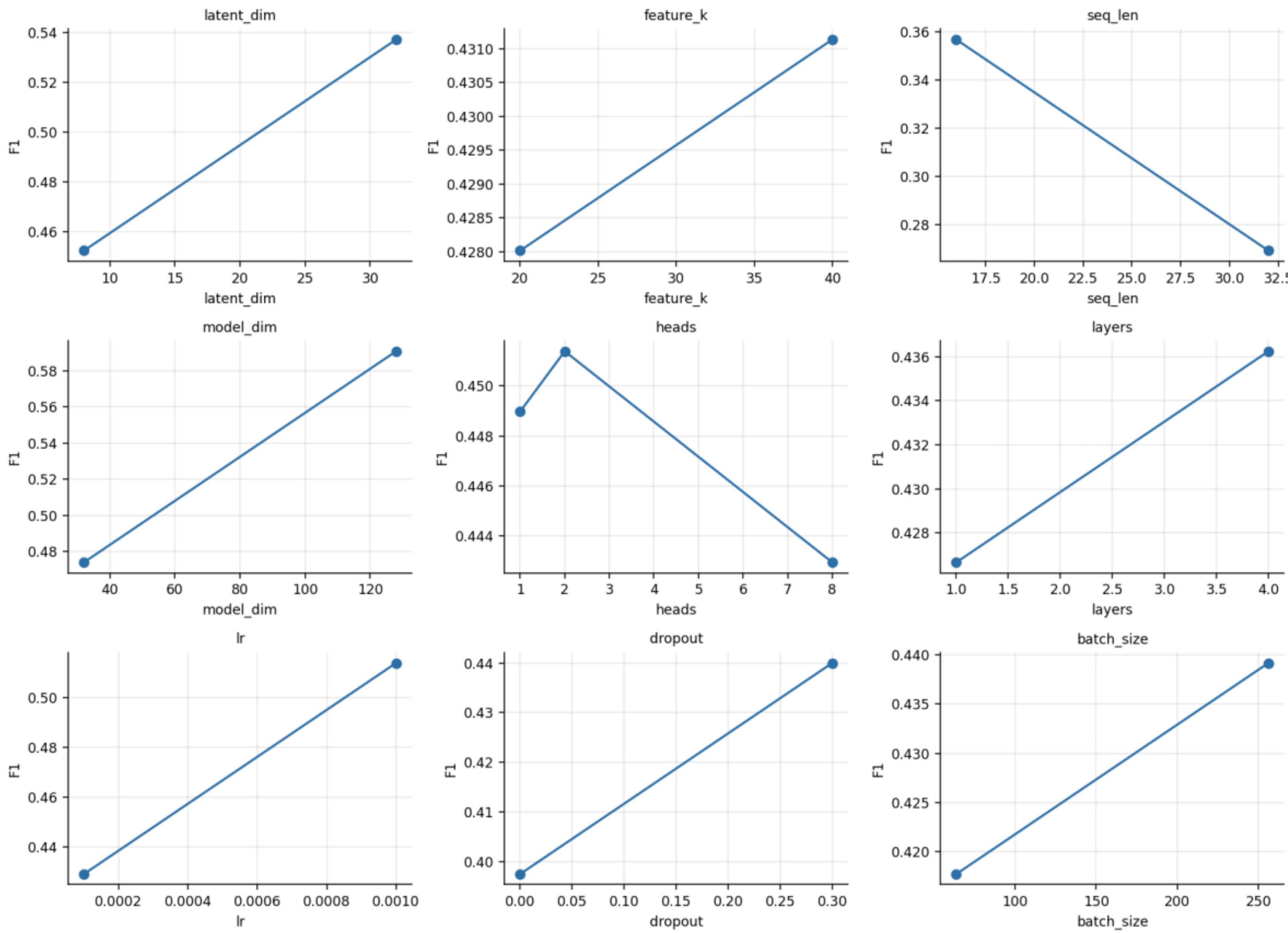
heatmap



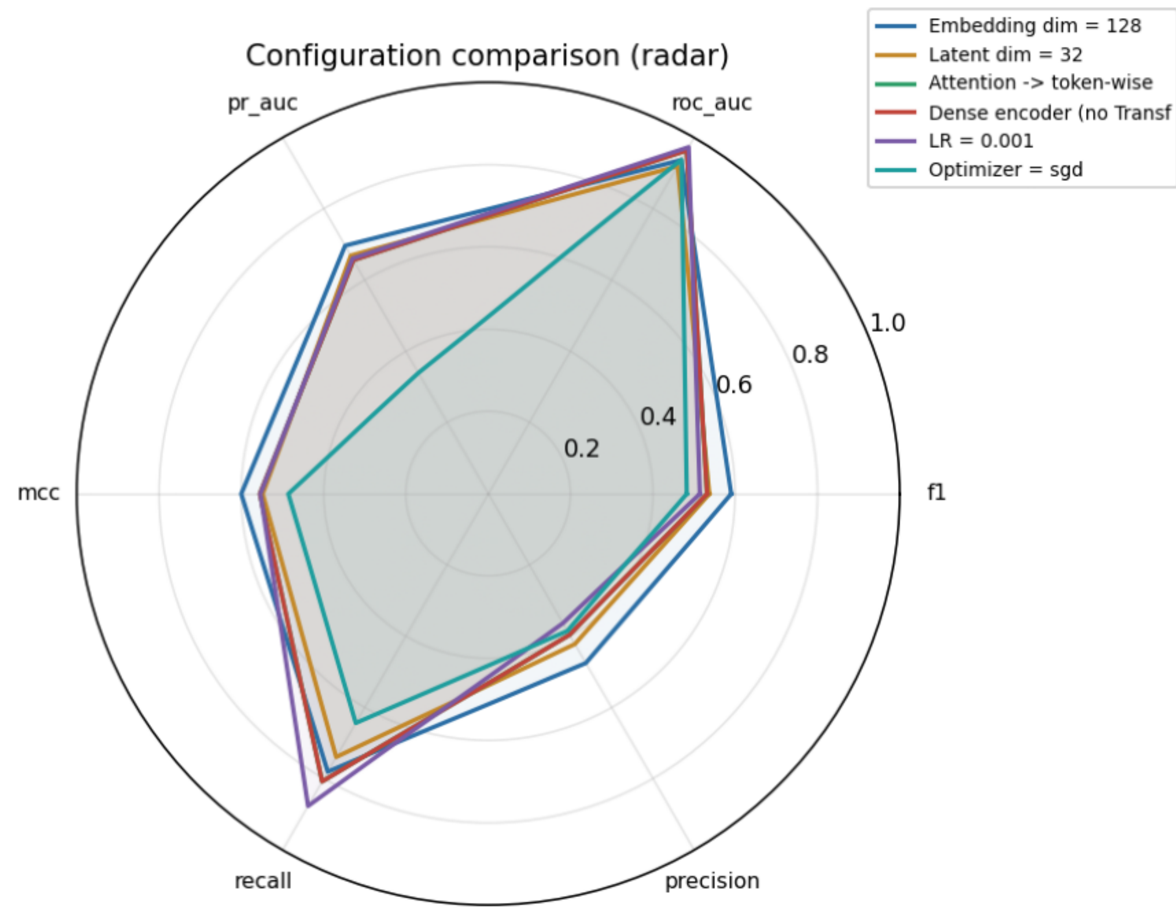
Ablation metric heatmap — test
Transformer + Autoencoder (baseline)
Dense encoder (no Transformer)
No latent bottleneck
No positional encoding
Attention -> token-wise FFN
Latent dim = 32
Latent dim = 8
Loss = HUBER
Loss = MAE
Threshold = adaptive
Threshold = mad
Threshold = p95
Threshold = p99
Top-20 features (variance)
Top-40 features (variance)
No normalization (raw features)
Sequence length = 16
Sequence length = 32
Embedding dim = 128
Embedding dim = 32
Heads = 1
Heads = 2
Heads = 8
Layers = 1
Layers = 4
LR = 0.0001
LR = 0.001
Optimizer = adam
Optimizer = rmsprop
Optimizer = sgd
Dropout = 0.0
Dropout = 0.3
Batch size = 256
Batch size = 64
Activation = elu
Activation = leaky_relu
Activation = relu
Without XAI attribution
accuracy
precision
recall
f1
roc_auc
pr_auc
mcc
balanced_accuracy
1.0
0.8
0.6
0.4
0.2
0.0

sensitivity

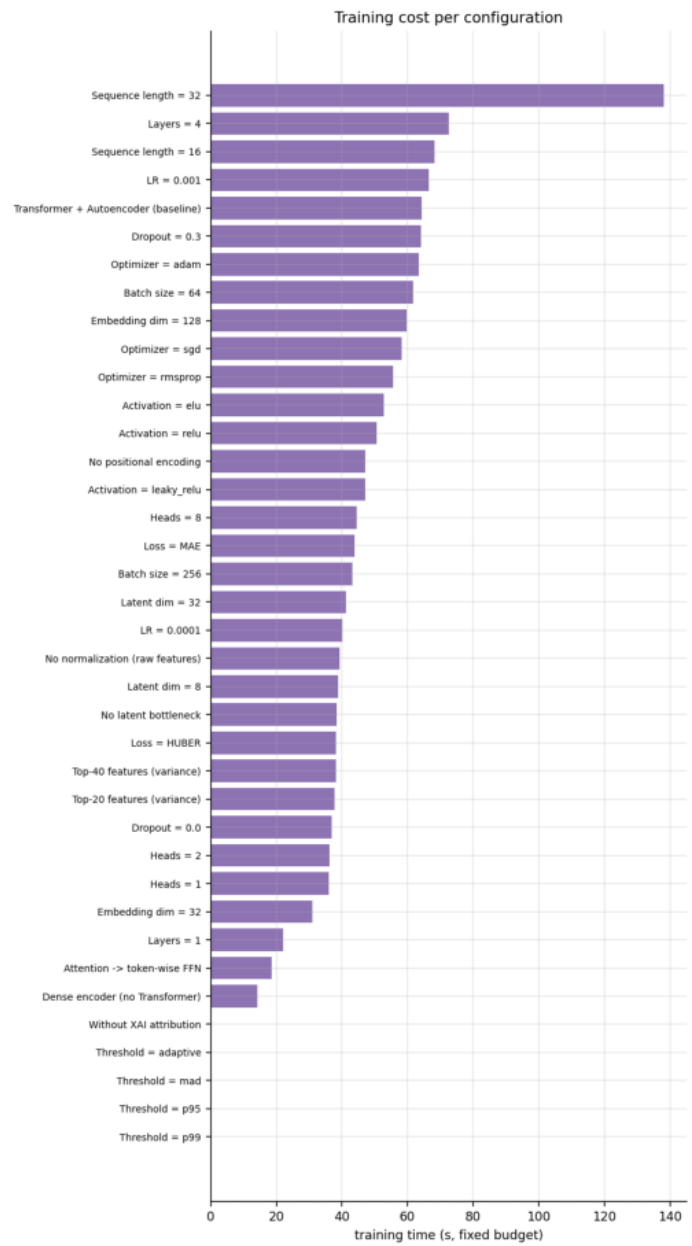
Hyperparameter sensitivity (F1 vs value)



radar


Configuration comparison (radar)
roc_auc
pr_auc
mcc
recall
precision
f1
0.2
0.4
0.6
0.8
1.0
Embedding dim = 128
Latent dim = 32
Attention -> token-wise
Dense encoder (no Transf
LR = 0.001
Optimizer = sgd

train_time


Training cost per configuration
Sequence length = 32
Layers = 4
Sequence length = 16
LR = 0.001
Transformer + Autoencoder (baseline)
Dropout = 0.3
Optimizer = adam
Batch size = 64
Embedding dim = 128
Optimizer = sgd
Optimizer = rmsprop
Activation = elu
Activation = relu
No positional encoding
Activation = leaky_relu
Heads = 8
Loss = MAE
Batch size = 256
Latent dim = 32
LR = 0.0001
No normalization (raw features)
Latent dim = 8
No latent bottleneck
Loss = HUBER
Top-40 features (variance)
Top-20 features (variance)
Dropout = 0.0
Heads = 2
Heads = 1
Embedding dim = 32
Layers = 1
Attention -> token-wise FFN
Dense encoder (no Transformer)
Without XAI attribution
Threshold = adaptive
Threshold = mad
Threshold = p95
Threshold = p99
0
20
40
60
80
100
120
140
training time (s, fixed budget)